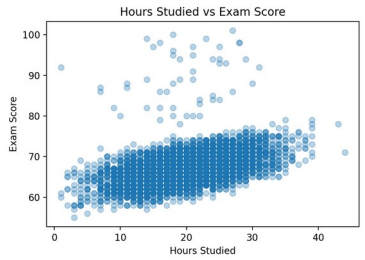
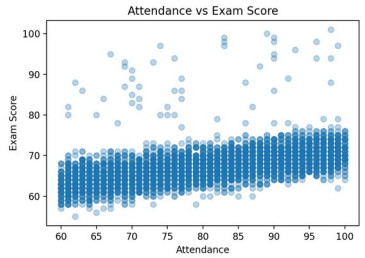
Student Performance Risk Classifier

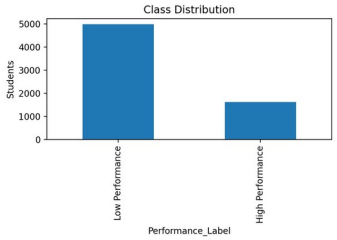
Final Project Nalar Bootcamp - klasifikasi performa siswa berbasis machine learning dengan deployment Streamlit.

Presenter: Anta

Dataset: StudentPerformanceFactors.csv (6,607 rows)
Target: High Performance jika Exam_Score >= 70







Background, Goal, dan Dampak

Masalah pendidikan yang diangkat dan alasan project dibuat

Background

• Sekolah sering terlambat mengenali siswa berisiko.
• Faktor seperti jam belajar, kehadiran, nilai sebelumnya, dan sesi tutoring dapat menjadi sinyal awal.
• Dibutuhkan sistem yang cepat, terukur, dan mudah diakses oleh guru maupun admin sekolah.

Rows	Columns	Target
6,607	20	Binary

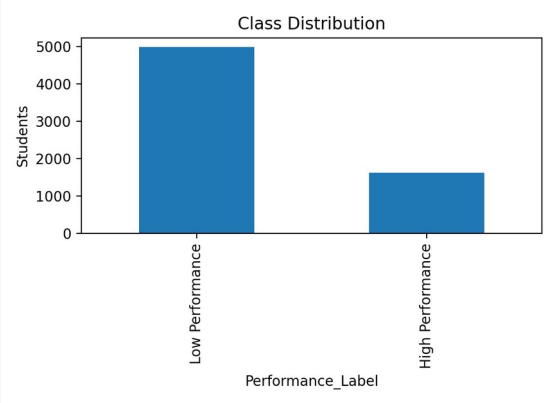
Goal & Impact

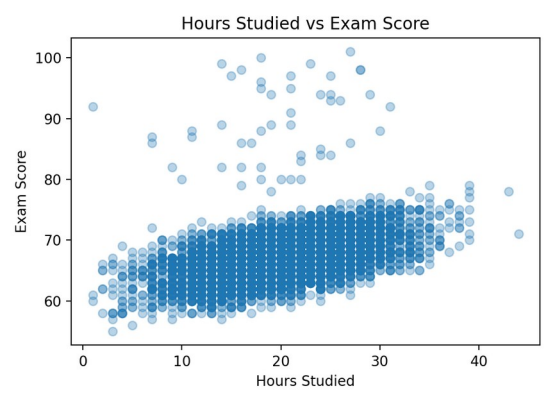
• Mengklasifikasikan siswa menjadi High vs Low Performance.
• Memberi dasar intervensi akademik lebih awal.
• Menyediakan dashboard analitik dan aplikasi prediksi yang dapat di-deploy ke Streamlit.

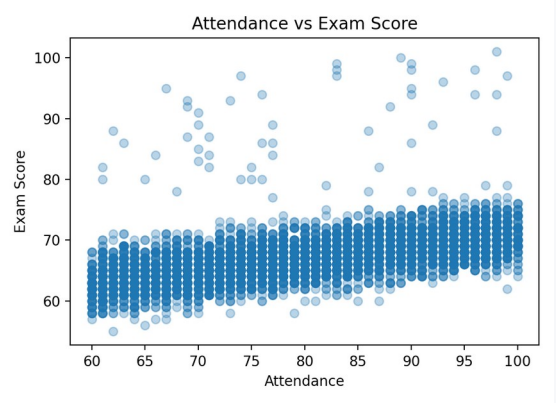
Accuracy	Precision	ROC-AUC
0.979	0.977	0.984

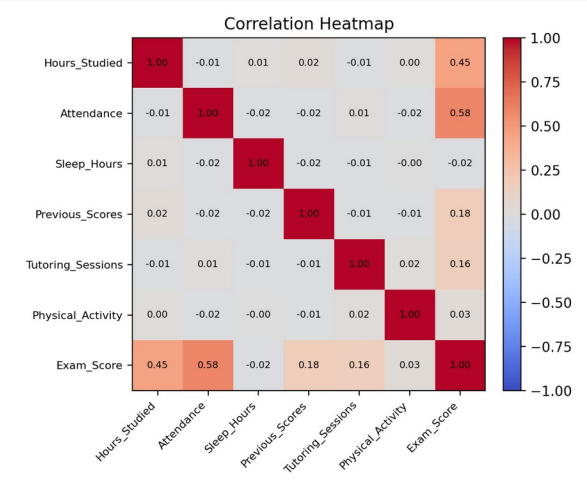
Dataset dan EDA

Distribusi kelas, hubungan fitur utama, dan korelasi numerik









Insight utama

- Distribusi target cukup layak untuk klasifikasi.
- Hours Studied dan Attendance memiliki hubungan positif terhadap Exam Score.
- Previous Scores dan dukungan belajar juga berkontribusi kuat.
- Hasil EDA dapat langsung dijadikan dashboard Tableau.

Flow Project

Alur kerja dari data mentah hingga deployment

1
Load Data
CSV 6,607 baris

2
Cleaning
Impute missing value

3
EDA
Plot + insight + Tableau

4
Targeting
Exam_Score >= 70

5
Modeling
4 model dibandingkan

6
Evaluation
Accuracy, Precision, Recall, ROC-AUC

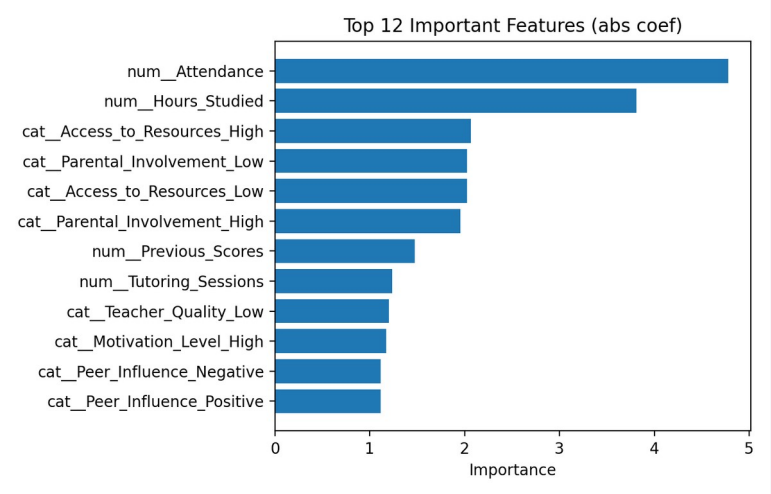
7
Deployment
Save pipeline + Streamlit

Deliverable akhir: model tersimpan, evaluasi, dashboard Tableau-ready, aplikasi Streamlit, Dockerfile, dan deck presentasi.

Model Comparison

Empat model diuji untuk memilih model produksi terbaik

Model	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression	0.979	0.977	0.935	0.984
Gradient Boosting	0.933	0.907	0.809	0.969
Random Forest	0.891	0.955	0.585	0.966
Neural Network (MLP)	0.966	0.938	0.923	0.984

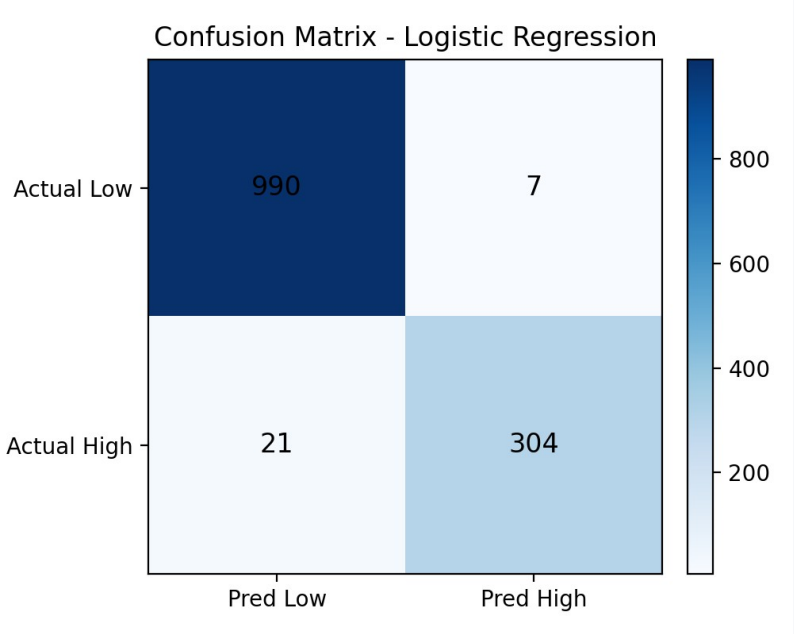


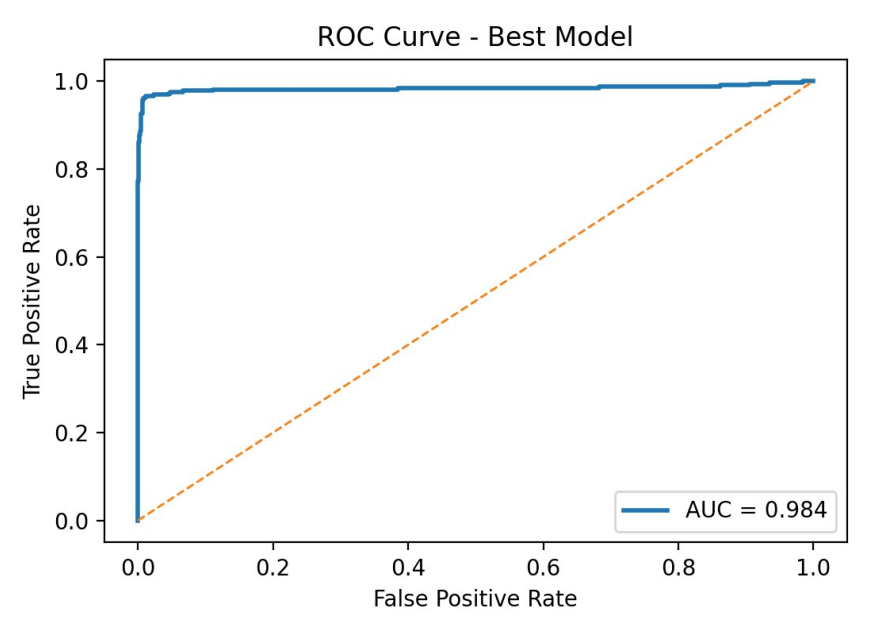
Model terbaik dipilih berdasarkan ROC-AUC dan stabilitas metrik. Logistic Regression unggul tipis atas MLP dan paling mudah diinterpretasi.

Best Model Evaluation

Best model: Logistic Regression

Accuracy	Precision	Recall	F1-Score	ROC-AUC
0.979	**0.977**	**0.935**	**0.956**	**0.984**





Implementation dan Deployment

Bagaimana sistem dibangun dan dijalankan

Tech stack

Python, pandas, scikit-learn, joblib, Streamlit, Docker, Tableau-ready CSV.

```
streamlit run app/streamlit_app.py
python -m src.train_model
```

Deployment target: Streamlit Community Cloud
Containerization: Dockerfile ready

Demo Time & Penutup

Apa yang bisa ditunjukkan saat presentasi 15 menit

Demo script

1. Tunjukkan struktur project
2. Jalankan training model
3. Buka app Streamlit
4. Input profil siswa
5. Tampilkan probabilitas high performance
6. Upload CSV untuk batch prediction

Kesimpulan

Project ini memenuhi kebutuhan final project: berbasis machine learning, memiliki EDA, siap untuk Tableau, deployment Streamlit, serta evaluasi model di atas ambang tugas. Paket ZIP sudah menyertakan kode, model, deck presentasi, dan Dockerfile.

Thank you